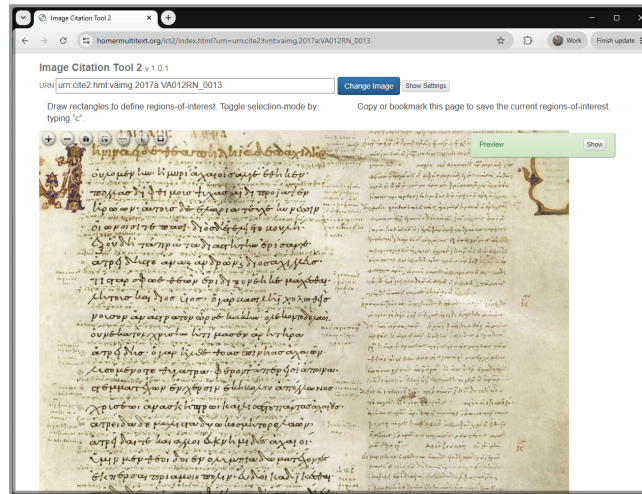


**What can we see
when we can see
overlap?**



ΡΑΨΩΔΙΑ Α

μήνιν ᾕειδε θεὰ Πηληϊάδεω Ἀχιλῆος
οὐλομένην, ἣ μυρ' Ἀχαιοῖς ἄλγε' ἔθηκε,
πολλὰς δ' ἰφθίμους ψυχὰς Ἄϊδι προΐαψεν
ἡρώων, αὐτοὺς δὲ ἑλώρια τεύχε κύνεσσιν
οἰωνοῖσι τε πᾶσι, Διὸς δ' ἐτέλεieto βουλῇ,
ἔξ οὗ δὴ τὰ πρῶτα διαστήτην ἐρίσαντε
Ἀτρεΐδης τε ἄναξ ἀνδρῶν καὶ δῖος Ἀχιλλεύς.
τίς τ' ἄρ σφωε θεῶν ἑριδι ζυνέθηκε μάχεσθαι;
Λητοῦς καὶ Διὸς υἱός· ὃ γὰρ βασιλῆϊ χολωθεὶς
νοῦσον ἀνὰ στρατὸν ὄρσε κακῆν, δλέκοντο δὲ λαοί,
οὐνεκα τὸν Χρῦσσην ἠτίμασεν Ἀρητῆρα
Ἀτρεΐδης· ὃ γὰρ ἦλθε θοὰς ἐπὶ νῆας Ἀχαιῶν
λυσόμενος τε θυγάτρα φέρων τ' Ἀπερείσι' ἄποινα,
στέμματα' ἔχων ἐν χερσίν ἐκπιβόλου Ἀπόλλωνος
χρυσῶν ἀνὰ σκήπτρῳ, καὶ λίσσεται πάντας Ἀχαιοὺς,
Ἀτρεΐδα δὲ μάλιστα δῶμα, κοσμήτορε λαῶν.
Ἀτρεΐδαι τε καὶ ἄλλοι εὐκνήμιδες Ἀχαιοί,
ὕμιν μὲν θεοὶ δοῖεν Ὀλύμπια δώματ' ἔχοντες
ἐκπέρασσι Πριάμοιο πόλιν, εὐ δ' οἴκαδ' ἰκέσθαι·
παῖδα δ' ἐμοὶ λύσασαι φίλῃν, τὰ δ' ἄποινα δέχεσθαι,
ἄζόμενοι Διὸς υἱὸν ἐκπιβόλον Ἀπόλλωνα.
ἐνθ' ἄλλοι μὲν πάντες ἐπευφήμησαν Ἀχαιοὶ
αἰδεῖσθαι θ' ἱερῆα καὶ ἀγλαὰ δέχθαι ἄποινα·
ἀλλ' οὐκ Ἀτρεΐδῃ Ἀγαμέμνονι ἦνδανε θυμῷ,
ἀλλὰ κακῶς ἄφίει, κρατερὸν δ' ἐπὶ μῦθον ἔτελλε·
μή σε γέρον κοίλῃσιν ἐγὼ παρὰ νηυσὶ κειμήλιον
ἦ νῦν δηθύνοντες ἢ ὕπερον αὐτὶς ἰόντα,
μή νύ τοι οὐ χροάσῃ σκήπτρον καὶ στέμμα θεοῖο·
τήν δ' ἐγὼ οὐ λύσω· πρὶν μιν καὶ γῆρας ἔπεισιν
ἡμετέρῳ ἐνὶ οἴκῳ ἐν Ἀργεῖ τηλόθι πάτρης
ἰστὸν ἐποιοχμένην καὶ ἐμόν λῆχος ἀντίοωσαν·
ἀλλ' ἴθι μὴ μ' ἐρέθιζε σῶαυτερος ὥς κε νέηαι.
ὥς ἔφατ', ἔδεισεν δ' ὃ γέρον καὶ ἐπείθετο μῦθῳ·
βῆ δ' ἀέκων παρὰ θῖνα πολυφλοίσβοιο θαλάσσης·
πολλὰ δ' ἔπειτ' ἀπάνευθε κιῶν ἦρθ' ὃ γεραιὸς
Ἀπόλλωνι ἄνακτι, τὸν ἡὔκομος τέκε Λητώ·
κλυθὶ μεν ἄργυρότοξ', ὃς Χρῦσσην ἀμφιβέβηκας
Κίλλαν τε Ζαθέην Τενεδίου τε ἴφι ἀνάσσεις,
Σμινθεὺ εἰ ποτέ τοι χαρίεντ' ἐπὶ νηὸν ἔρεψα,
ἦ εἰ δὴ ποτέ τοι κατὰ πῖονα μηρ' ἔκηα
ταύρων ἢ δ' αἰγῶν, τὸ δέ μοι κρήνην ἐέλδωρ·
τίσειαν Δαναοὶ ἐμὰ δάκρυα σοῖσι βέλεσσιν.
ὥς ἔφατ' εὐχόμενος, τοῦ δ' ἔκλυε Φοῖβος Ἀπόλλων,
βῆ δὲ κατ' Οὐλύμπιοι καρήνων χωόμενος κῆρ,
τόξ' ὤμοισιν ἔχων ἀμφηρεφέα τε φαρέτηρην·
ἐκλαγξαν δ' ἄρ' οἴστοι ἐπ' ὤμων χωομένοιο,
αὐτοῦ κινηθέντος· ὃ δ' ἦτε νυκτὶ εἰοικώς.
ἔζειτ' ἔπειτ' ἀπάνευθε νεῶν, μετὰ δ' ἰὸν ἔθηκε·
δεινὴ δὲ κλαγγὴ γένετ' ἄργυρέοιο βιοῖο·
οὐρήας μὲν πρῶτον ἐπώχeto καὶ κύνας ἄργούς,
αὐτὰρ ἔπειτ' αὐτοῖσι βέλος ἐχευευκὲς ἐφίεις
βάλλ'· αἰεὶ δὲ πυρὰ νεκρῶν καίοντο θαμέαι.



Screenshot from Homer Multitext Project
Showing a page from the 10th century CE (Iliad Venetus A)
<https://www.homer-multitext.org/ict2/index.html>

Topic of interest:
Closely reading Homer's *Iliad*

[BOOK [n]1{}}]

[para][prayer [to]Calliope{}][1 [n]1.1{]}μήνιν ἄειδε θεὰ Πηληϊάδεω Ἀχιλῆος{1}
[1 [n]1.2{]}οὐλομένην, ἣ μυρί' Ἀχαιοῖς ἄλγε' ἔθηκε,{1}
[1 [n]1.3{]}πολλὰς δ' ἰφθίμους ψυχὰς Ἄϊδι προΐαψεν{1}
[1 [n]1.4{]}ἡρώων, αὐτοὺς δὲ ἑλώρια τεῦχε κύνεσσιν{1}
[1 [n]1.5{]}οἰνονοῖσιν τε πᾶσι, Διὸς δ' ἐτελείετο βουλή,{1}
[1 [n]1.6{]}ἔξ οὗ δὴ τὰ πρῶτα διαστήτην ἐρίσαντε{1}
[1 [n]1.7{]}Ἀτρεΐδης τε ἀναξ ἀνδρῶν καὶ δῖος Ἀχιλλεύς,{1}[prayer][para]
[para][1 [n]1.8{]}τίς τ' ἄρ' σφωε θεῶν ἑριδοῖ ξυνέηκε μάχεσθαι,{1}
[1 [n]1.9{]}Ἀητοὺς καὶ Διὸς υἱός· ὁ γὰρ βασιλῆϊ χολωθεὶς{1}
[1 [n]1.10{]}νοῦσον ἀνὰ στρατὸν ὄρσε κακὴν, ὀλέκοντο δὲ λαοί,{1}
[1 [n]1.11{]}οὐνεκα τὸν Χρῦσῃν ἠτίμασεν Ἀρητὴρα{1}
[1 [n]1.12{]}Ἀτρεΐδης· [scene [set]Achaian ships{}]ὁ γὰρ ἦλθε θοὰς ἐπὶ νῆας Ἀχαιῶν{1}
[1 [n]1.13{]}λυσόμενός τε θύγατρα φέρων τ' ἄπερείσι' ἄποινα,{1}
[1 [n]1.14{]}στέμματα' ἔχων ἐν χερσὶν ἐκηβόλου Ἀπόλλωνος{1}
[1 [n]1.15{]}χρυσῆν ἀνὰ σκήπτρῳ, καὶ λίσσετο πάντας Ἀχαιοὺς,{1}
[1 [n]1.16{]}Ἀτρεΐδα δὲ μάλιστα δῶν, κοσμήτορε λαῶν·{1}
[speech [by]Chryses{}][to]Agamemnon, Menelaus, Achaians{}
[1 [n]1.17{]}Ἀτρεΐδαι τε καὶ ἄλλοι ἐϋκνήμιδες Ἀχαιοί,{1}
[1 [n]1.18{]}ὕμιν μὲν θεοὶ δοῖεν Ὀλύμπια δώματ' ἔχοντες{1}
[1 [n]1.19{]}ἐκπέρσαι Πριάμοιο πόλιν, εὖ δ' οἴκαδ' ἰκέσθαι·{1}
[1 [n]1.20{]}παῖδα δ' ἐμοὶ λύσαιτε φίλην, τὰ δ' ἄποινα δέχεσθαι,{1}
[1 [n]1.21{]}ἄζόμενοι Διὸς υἱὸν ἐκηβόλον Ἀπόλλωνα·{1}[speech][para]
[para][1 [n]1.22{]}ἐνθ' ἄλλοι μὲν πάντες ἐπευφήμησαν Ἀχαιοί·{1}
[1 [n]1.23{]}αἰδεῖσθαι θ' ἱερῇα καὶ ἀγλαὰ δέχθαι ἄποινα·{1}
[1 [n]1.24{]}ἀλλ' οὐκ Ἀτρεΐδῃ Ἀγαμέμνονι ἦνδανε θυμῷ,{1}
[1 [n]1.25{]}ἀλλὰ κακῶς ἄφειλ, κρατερὸν δ' ἐπὶ μῦθον ἔτελλε·{1}
[speech [by]Agamemnon{}][to]Chryses{}
[1 [n]1.26{]}μή σε γέρον κοίλῃσιν ἐγὼ παρὰ νηυσὶ κιχέω{1}
[1 [n]1.27{]}ἢ νῦν δηθύνοντ' ἢ ὕστερον αὖτις ἰόντα,{1}
[1 [n]1.28{]}μή νύ τοι οὐ χραίσμη σκήπτρον καὶ στέμμα θεοῖο·{1}
[1 [n]1.29{]}τὴν δ' ἐγὼ οὐ λύσω· πρὶν μιν καὶ γῆρας ἔπεισιν{1}
[1 [n]1.30{]}ἡμετέρῳ ἐνὶ οἴκῳ ἐν Ἀργεῖ τηλόθι πάτρης{1}
[1 [n]1.31{]}ἰστον ἐπιτοχομένην καὶ ἐμὸν λῆχος ἀντιώσωσαν·{1}
[1 [n]1.32{]}ἀλλ' ἴθι μὴ μ' ἐρέθιζε σῴτερος ὥς κε νέηαι·{1}[speech][para]
[para][1 [n]1.33{]}ὥς ἔφατ', ἔδειξεν δ' ὁ γέρον καὶ ἐπείθετο μῦθῳ·{1}[scene]
[scene [set]seashore{}][1 [n]1.34{]}βῆ δ' Ἀχαιῶν παρὰ θῖνα πολυφλοίσβοιο θαλάσσης·{1}
[1 [n]1.35{]}πολλὰ δ' ἔπειτ' ἀπάνευθε κιὼν ἡρᾷθ' ὁ γεραιὸς{1}
[1 [n]1.36{]}Ἀπόλλωνι ἄνακτι, τὸν ἡὔκομος τέκε Λητώ·{1}
[prayer [to]Apollo{}][by]Chryses{}
[1 [n]1.37{]}κλυθὶ μευ ἀργυρότοξ', ὃς Χρῦσῃν ἀμφιβέβηκας{1}

What are we trying to achieve? Exploratory Markup

Unrealized potentials
of declarative text encoding
Describing data with tags
regarding overlap as normal
... We can tag freely as we go ...
A schema is an object of study
(not a binding commitment)
"More like guidelines"
Markup first, hierarchies later

Tools use **Laminator** XProc pipelines under XML Calabash or Morgana-IIIse (both Java)

Tagging can be even more descriptive

... and a SaxonJS-based platform is contemplated

What if XML had no end-tag matching rule?



Laminator

<https://github.com/wendellpiez/Laminator>

How it works

An Invisible XML grammar plus handful of XSLTs

Glued together with XProc+iXML

LMNL syntax is read in via iXML parse plus XSLT x 2

XML can be ingested directly, or convert to LMNL

Results can be expressed in LMNL or XML syntax

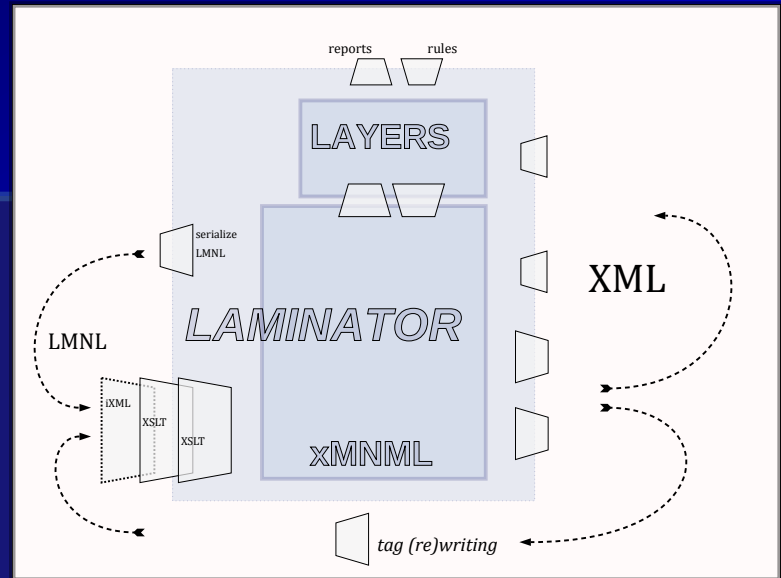
Works with any XML idiom for expressing overlap

Data flows are adaptable

problem domain

data set

tagging semantics



LMNL inputs

Made by hand

Or generated from (XML or other) source

[line [no]5.791{ }]ώς ἔφατ', [l1tt]οὐ δ' ἀπίθησε [nom]θεὰ[l1tt] γλαυκῶπις Ἀθήνη[nom].{line]

xMNML - a tagged text (sequence) model

```
PS C:\Users\wapie\Documents\Projects\Github\EpicMarkup\run> xc .\RUN-ONE-LINE_LMNL.xml
Invoking XML Calabash 3.0.54 for XProc 3 pipeline execution...
=====
== xMNML :: 1 ::
<?xml version="1.0" encoding="UTF-8"?>
<LMNL xmlns="http://wendellpiez.com/ns/xMNML">
  <start gi="line" rID="r1_line" L="1" ch="1" off="0" ext="44">
    <annotation L="1" ch="7" gi="no">5.791</annotation>
  </start>
  <text cf="r1_line" off="0" ext="10">ὥς ἔφατ', </text>
  <start gi="l1tt" rID="r2_ltt" L="1" ch="20" off="18" ext="17"/>
  <text cf="r1_line r2_ltt" off="18" ext="10">οὐ δ' ἀπίθησε </text>
  <start gi="nom" rID="r3_nom" L="1" ch="48" off="24" ext="19"/>
  <text cf="r1_line r2_ltt r3_nom" off="24" ext="3">θεὰ</text>
  <end gi="l1tt" rID="r2_ltt" L="1" ch="56" off="27" ext="17"/>
  <text cf="r1_line r3_nom" off="27" ext="16">γλαυκῶπις Ἀθήνη</text>
  <end gi="nom" rID="r3_nom" L="1" ch="77" off="43" ext="19"/>
  <text cf="r1_line" off="43" ext="1">.</text>
  <end gi="line" rID="r1_line" L="1" ch="83" off="44" ext="44"/>
</LMNL>
=====
Elapsed time: 00:00:01.9967690
Press any key to continue...
```



LAYERS - a range model

```
PS C:\Users\wapie\Documents\Projects\Github\EpicMarkup\run> xc .\RUN-ONE-LINE_LMNL.xml
Invoking XML Calabash 3.0.54 for XProc 3 pipeline execution...
=====
== LAYERS :: 1 ::
<?xml version="1.0" encoding="UTF-8"?>
<LAYERS xmlns="http://wendellpiez.com/ns/xMNML">
  <layer ranges="line ltt nom" fore="ech">
    <range rID="r1_line" gi="line" starting="0" ending="44">
      <annotation L="1" ch="7" gi="no">5.791</annotation>
    </range>
    <range rID="r2_ltt"
      gi="l1tt"
      starting="10"
      ending="27"/>
    <range rID="r3_nom"
      gi="nom"
      starting="24"
      ending="43"
      overlaps="r2_ltt"/>
  </layer>
  <frontier>ὥς ἔφατ', οὐ δ' ἀπίθησε θεὰ γλαυκῶπις Ἀθήνη.</frontier>
</LAYERS>
=====
Elapsed time: 00:00:01.9888409
Press any key to continue...
```

XML results

... one (any) hierarchy at a time

<line no="5.791">ὥς ἔφατ', οὐ δ' ἀπίθησε <nom>θεὰ γλαυκῶπις Ἀθήνη</nom>.</line>

<line no="5.791">ὥς ἔφατ', <l1tt>οὐ δ' ἀπίθησε θεὰ</l1tt> γλαυκῶπις Ἀθήνη.</line>

<line no="5.791">ὥς ἔφατ', <l1tt>οὐ δ' ἀπίθησε <nom>θεὰ</nom></l1tt><nom> γλαυκῶπις Ἀθήνη</nom>.</line>

<line no="5.791">ὥς ἔφατ', <l1tt_s>οὐ δ' ἀπίθησε <nom>θεὰ<l1tt_e/> γλαυκῶπις Ἀθήνη</nom>.</line>

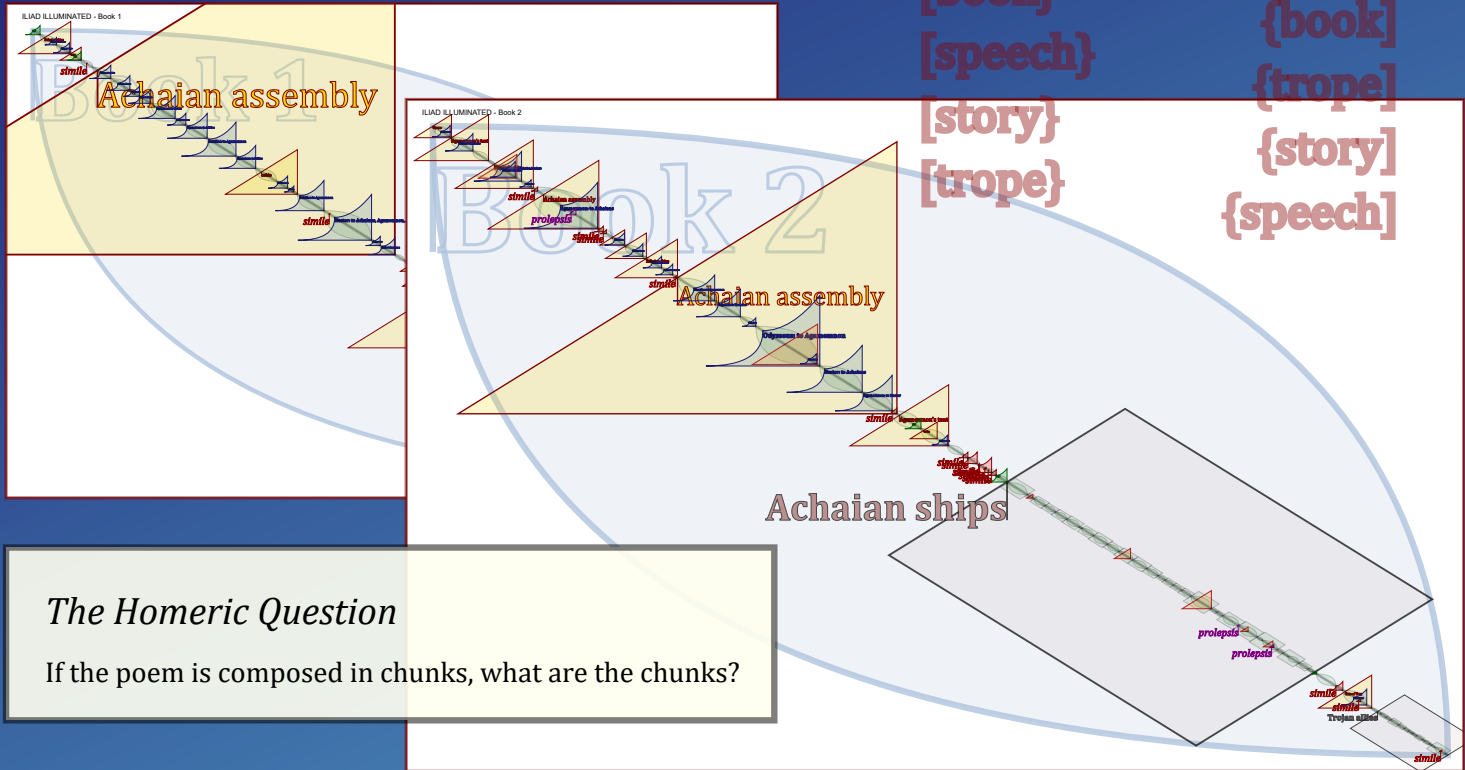
Epic Findings

The fractal structure of Homer's *Iliad*

Units of measure

morpheme, word (ονομα), line (στιχος), phalanx (φαλαγγ: ~10 lines), stratos (στρατος: ~100 lines), rhapsody (ραψωδια ~1000 lines)

Form and formula (*all the way down*) — WAVE or KYMA (κυμα) pattern



The Homeric Question

If the poem is composed in chunks, what are the chunks?

Range maps depict the extent of ranges of text within the poem.

Project goals and progress

<https://github.com/wendellpiez/EpicMarkup>

<https://github.com/wendellpiez/Laminator>

Free to inspect, assess, acquire, apply, adapt



Technical topics

Markup, tags, hierarchies and schemas

Whitespace (always)

Tag (re)writing vs/and transformation

Why XProc is nice for this

Parallel initiatives (by the developer)

Using public domain data from PerseusDL and Homer Multitext Project

Homer helper site (Scriptorium Method):

<https://raventracks.org/>

Lightweight *Iliad* Assisted Reader: **Iliad Elucidated**

https://raventracks.org/#elucidated_iliad

LMNL, MNML, Laminator

LMNL markup is now easy to make

From XML or from scratch

More data, more examples?

Applications: narrative and genre studies, textual criticism and others

TODO: Formal specifications for MNML LMNL as implemented

TODO: Laminator documentation and testing

TODO: Alternate implementations including XPath 4.0 (record objects), All-XSLT/iXML, no-XML?

TODO: LMNL Annotations; Hyperdocument model; XML Profiles; DTS 1.0 API ...?

To come: **Iliad Illuminated**

(Some day, by small steps)

Awaiting findings first

Ideas are welcome!

For now - range map diagrams and reading notes